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Joint Channel-and-Architecture Search for EEG Classification of Neurological Disorders

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ABSTRACT

Neural Architecture Search (NAS) is a promising strategy for automated model design in electroencephalography (EEG), but clinical adoption remains constrained by limited sample sizes, strong inter-subject variability, and hardware constraints in wearable settings. This work presents a differentiable joint optimization framework that learns EEG channel subsets and neural architectures simultaneously under a multiobjective criterion balancing predictive performance, sensor burden, and structural cost. Experiments were conducted on three public datasets (depression: $N = 119$, $C = 19$; schizophrenia: $N = 84$, $C = 16$; epilepsy: $N = 318$, $C = 14$) using stratified 5-fold cross-validation and a three-phase training schedule (20/70/20 nominal epochs with phase-wise early stopping). Relative to fixed-architecture and NAS baselines, the proposed method achieved consistent channel reduction (68–71%) while maintaining competitive macro-F_1 scores (depression: 0.778 ± 0.075, schizophrenia: 0.723 ± 0.089, epilepsy: 0.602 ± 0.074). The strongest gain was observed for schizophrenia, where joint optimization outperformed all reference deep baselines under severe data scarcity. The reported ablations show that removing channel selection increases F_1 but requires substantially more electrodes, illustrating a clinically relevant Pareto trade-off between diagnostic performance and acquisition burden. The reported stability analysis further indicates that channel selection is more consistent than operation selection across the independent runs considered here, with the highest operation consistency observed in the largest cohort. These findings support joint channel-and-architecture optimization as a practical route toward compact, interpretable, and deployable EEG diagnostic systems.

INDEX TERMS EEG, neural architecture search, channel selection, assistive diagnosis, neurological disorders, embedded inference

I. INTRODUCTION

Neural Architecture Search (NAS) has emerged as a major paradigm shift in deep learning by automating the design of neural architectures, consistently surpassing manually engineered models across vision, mobile computing, and natural language processing benchmarks [3]. Representative studies on large-scale image classification and resource-constrained deployment have shown that systematic exploration of the architectural search space can uncover solutions beyond human intuition [2]. In parallel, methodological advances have substantially reduced the computational cost of NAS, decreasing requirements from thousands of GPU-days to levels accessible in academic and clinical research settings [15].

Despite these advances, the adoption of NAS in clinical applications involving biomedical time-series remains limited [14]. Unlike vision or language domains, physiological signals commonly involve small datasets, high inter-subject variability, strongly correlated multi-channel measurements, and demanding interpretability requirements in clinical use [18]. Under these conditions, NAS approaches originally developed for large-scale data and homogeneous architectural spaces become inefficient and poorly aligned with clinical realities.

Electroencephalography (EEG) exemplifies these challenges. As a core modality for diagnosing and monitoring neurological and psychiatric disorders affecting hundreds of millions of individuals worldwide [4], EEG provides a

temporally rich representation of brain dynamics, combining high sampling rates with spatially distributed channels [7]. However, this richness introduces severe combinatorial complexity, as even moderate EEG configurations with 14 channels yield 2^{14} possible channel subsets, which further multiply when combined with heterogeneous architectural paradigms such as convolutional, recurrent, and attention-based models [11].

Beyond combinatorial growth, clinical EEG analysis faces three additional structural barriers. First, the computational cost of traditional NAS is incompatible with clinical small-data regimes, where datasets often include only a few hundred patients [8]. Second, there is no architectural consensus across clinical tasks, with models effective for epilepsy, depression, or schizophrenia differing substantially in structure and inductive bias [5]. Third, clinical deployment requires neurophysiological interpretability, such that selected channels and temporal patterns must correspond to meaningful anatomical brain regions, a requirement largely ignored in conventional NAS pipelines [5].

Early studies applying NAS to EEG reported performance gains in non-clinical domains such as motor imagery and emotion recognition [16]. However, these approaches share critical limitations, including high computational cost, lack of explicit channel selection, restriction to a single architectural search space, and validation on tasks that do not reflect real-world clinical constraints [13]. As a result, existing EEG-NAS methods remain distant from translational clinical use.

In parallel, the emergence of zero-cost proxies has reshaped the NAS landscape by enabling architecture evaluation without full training, relying instead on gradient-based and synaptic-flow measures [9]. Methods based on connection sensitivity and information flow demonstrated strong rank correlation with fully trained performance while reducing evaluation time to milliseconds [10]. Despite their impact, zero-cost NAS methods remain largely unexplored in biomedical time-series, even as recent surveys explicitly call for efficient NAS solutions under clinical resource constraints.

Conversely, EEG channel selection has traditionally been treated as an independent problem, addressed through evolutionary algorithms or classical machine learning techniques [17]. While effective in reducing dimensionality, these methods require full model training for each candidate configuration, rendering joint optimization of architecture and channel selection computationally infeasible [6].

Consequently, a fundamental gap persists: the absence of a unified, computationally efficient, and clinically interpretable framework capable of jointly optimizing neural architectures and EEG channel subsets under severe data and resource constraints [1].

To address this gap, we propose SNIP-it, a differentiable multiobjective NAS framework that integrates architecture search and EEG channel pruning within a single optimization process. By coupling gradient-based channel saliency estimation to explicit sensor, MAC, and parameter penalties during search, the framework jointly learns compact channel masks and heterogeneous neural architectures under a unified clinical NAS paradigm [12], [19].

II. MATERIALS AND METHODS

A. END-TO-END OPERATIONAL ARCHITECTURE

The proposed framework is organized as a closed-loop methodological pipeline for assistive EEG analysis, integrating signal acquisition, compact representation learning, near-real-time inference, data persistence, model updating, and controlled deployment within a single operational cycle. This organization is intended to connect the full trajectory from incoming electrophysiological measurements to model revision, thereby aligning diagnostic support with longitudinal data accumulation and iterative performance improvement.

In the acquisition stage, multichannel EEG is processed through sliding temporal windows, which enables continuous monitoring while preserving temporal locality. Each window then undergoes a feature-extraction stage in which every channel is summarized by five canonical frequency bands combined with four attributes per band, yielding a compact representation of size $[C, 20]$. The resulting channel-by-feature tensor is normalized with z-score scaling to reduce inter-window variability. This design choice decreases inference cost relative to raw-signal processing while preserving clinically relevant spectral information for neurological classification tasks.

The inference stage operates on the compact representation and returns not only the predicted class, but also class probabilities, processing latency, and information about the active electrode configuration used by the model. This expanded output structure supports operational traceability and facilitates downstream auditing of performance, efficiency, and sensor utilization. To ensure service continuity, model access follows a hierarchical loading strategy in which the currently promoted registry version is prioritized while previously validated local artifacts remain available as fallback alternatives, thereby reducing the risk of interruption during model updates.

All processed windows and their associated annotations are stored in a persistent data layer that supports both auditability and future learning. Once labeled samples become available, they are incorporated into an automated updating pipeline that exports curated data, performs retraining and reevaluation, and promotes a new model version only when the target metric improves without violating cost or latency constraints. In this way, the framework implements a controlled form of continual improvement in which model evolution is conditioned on empirical evidence rather than unconditional replacement.

After a candidate model has been accepted, a deployment stage converts the learned solution into a compact inference-ready form suitable for runtime operation under hardware constraints. The promoted artifact is a slim discrete model accompanied by versioned metadata describing active channels

and input configuration. The production environment periodically checks for promoted versions and can replace the active model without interrupting the surrounding service. This end-to-end architecture is particularly relevant for assistive EEG diagnostics because it unifies data collection, efficient prediction, cost-aware deployment, and adaptive updating while preserving the interpretation that all predictions remain supportive rather than definitive clinical decisions.

B. MODEL DEVELOPMENT

This study formulates EEG classification as a joint optimization problem over channel-mask logits θ, backbone weights ϕ, and architectural logits α. The goal is to maximize macro-F_1 while minimizing sensor burden and structural cost under hardware-constrained inference.

Table 1 summarizes the three public datasets. Each sample is represented as a matrix $[C \times T]$, where $T = 20$ denotes the per-channel feature dimension obtained from five canonical EEG bands (delta, theta, alpha, beta, and gamma) combined with four attributes per band. This compact representation is specifically suited to low-power deployment because it avoids raw-signal preprocessing at inference time.

TABLE 1. EEG datasets used in this study.

Dataset	N	Channels (C)	Features (T)	Classes	Task
Depression	119	19	20	2	MDD vs. control
Schizophrenia	84	16	20	2	SZ vs. control
Epilepsy	318	14	20	2	Seizure vs. non-seizure

All comparative experiments reported in this manuscript were conducted using stratified 5-fold cross-validation with fixed split seed 42. Consistent with the project validation framework, the main comparative tables use this fold-based protocol, whereas stability assessment is treated as a separate multi-seed evidence block. Within each fold, models were trained with mini-batch size 8 using Adam (learning rate 10^{-3}, weight decay 10^{-4}) under a three-phase schedule comprising warm-up, joint search, and discrete fine-tuning. The nominal phase lengths were 20/70/20 epochs, with phase-wise early stopping (patience 15), best-state restoration, and Macro-F_1 as the primary selection metric.

TABLE 2. Training protocol.

Setting	Value
Cross-validation	Stratified 5-fold (split seed = 42)
Phase schedule	20/70/20 epochs (warm-up/search/fine-tune)
Batch size	8
Optimizer	Adam ($\eta = 10^{-3}$, weight decay $= 10^{-4}$)
Early stopping	Patience = 15, phase-wise best-state restore
Arch. LR multiplier	$10\times$
Default threshold	$\tau = 0.5$
Primary metric	Macro-F_1

To establish robust performance bounds, eleven baselines were evaluated, including fixed architectures, statistical channel selection, differentiable masking without architecture search, NAS with fixed channels, and EEG-specific deep networks (GRU, BiGRU, BiGRU-attention, and ConvNet). This design isolates the contribution of channel pruning from the contribution of architecture optimization.

Primary cross-dataset performance summaries are reported for the reference run (seed = 0) under the protocol above, whereas dedicated stability analyses follow the framework's multi-seed validation block.

C. DIFFERENTIABLE JOINT OPTIMIZATION OF CHANNELS AND ARCHITECTURE

The proposed search strategy treats electrode selection and neural architecture design as a single optimization problem, so that predictive performance, sensor burden, and computational efficiency are balanced throughout model development. For input $x \in \mathbb{R}^{C\times T}$ and label y, the joint objective is

$$\begin{aligned}\min_{\theta,\phi,\alpha}\ & \mathcal{L}_{task}(\theta,\phi,\alpha) + \lambda_{mac}\mathcal{C}_{mac}(\alpha) \\ & + \lambda_{sen}\mathcal{C}_{sen}(\theta) + \lambda_{par}\mathcal{C}_{par}(\alpha) \\ & + \mathcal{P}_{min}(\theta) + \mathcal{P}_{ent}(\theta)\end{aligned} \tag{1}$$

where θ denotes channel-mask logits, ϕ the backbone weights, and α the architectural logits. This formulation allows the representation induced by the retained EEG channels to influence architectural specialization during the same learning process, producing solutions that are simultaneously compact and task-adaptive.

The channel-selection stage assigns a differentiable relevance weight to each electrode through a sigmoid mask,

$$w_c = \sigma(\theta_c), \qquad \tilde{x}_{b,c,t} = w_c\, x_{b,c,t}, \tag{2}$$

where $\tilde{x}$ is the masked input tensor. Sensor burden is incorporated through a normalized cost term,

$$\mathcal{C}_{sen} = \frac{\sum_{c=1}^{C} w_c s_c}{\sum_{c=1}^{C} s_c}, \tag{3}$$

in which s_c can be uniform, channel-weighted, or region-weighted depending on the acquisition scenario.

Concurrently, the architectural search stage models each computational block as a probabilistic mixture of candidate operations. For cell l, the probability of operation i is

$$p_i^{(l)} = \mathrm{softmax}(\alpha^{(l)})_i, \tag{4}$$

and the default search space includes `conv1d_k3`, `conv1d_k5`, `dsconv1d`, `gru1d`, lightweight attention, and identity mappings. Pooling operators exist in the implementation registry but are not part of the default search space considered in the main configuration. Arithmetic and parameter costs are estimated in expectation as

$$\mathcal{C}_{mac} = \sum_{l=1}^{L}\sum_{i=1}^{|\mathcal{O}|} p_i^{(l)} m_i, \qquad \mathcal{C}_{par} = \sum_{l=1}^{L}\sum_{i=1}^{|\mathcal{O}|} p_i^{(l)} q_i \tag{5}$$

where m_i and q_i are the precomputed MAC and parameter counts of operation i.

To avoid degenerate sensor configurations, the objective includes two complementary regularizers over the channel mask,

$$\mathcal{P}_{min} = \gamma \sum_{k \in \text{top-}K(w)} \max(0, 0.5 - k)^2, \tag{6}$$

which discourages collapse toward nearly null channel sets, and

$$\mathcal{P}_{ent} = \frac{1}{C}\sum_{c=1}^{C} \left[-w_c \log(w_c) - (1 - w_c)\log(1 - w_c)\right], \tag{7}$$

which pushes the mask toward polarized values and reduces ambiguity during discretization.

Because clinical EEG datasets often operate under severe sample scarcity, the framework uses dataset-size-adaptive regularization coefficients,

$$\lambda_{eff} = \lambda \cdot \min\left(1, \frac{N}{N_{ref}}\right), \qquad N_{ref} = 1000, \tag{8}$$

so that smaller cohorts receive milder penalization than larger ones. In the phase-specific training objective below, these dataset-scaled terms are written explicitly through effective coefficients.

After convergence, the continuous solution is transformed into a deployable discrete model by retaining channel c when $\sigma(\theta_c) > 0.5$ and selecting, in each cell, the operation with maximum architectural logit. The resulting model therefore combines a reduced electrode set with an architecture specialized for the target disorder, yielding a clinically oriented compromise among diagnostic discrimination, acquisition practicality, and inference efficiency.

D. THREE-PHASE TRAINING PROTOCOL

To stabilize joint optimization in low-sample EEG scenarios, the training procedure is organized into three successive phases that separate representation consolidation, structural adaptation, and final calibration. This staged design reduces interference between predictive learning and compactness constraints, while improving consistency between the continuous search process and the discrete model used during deployment.

The first phase is devoted to representation warm-up and uses a nominal budget of 20 epochs. During this stage, the predictive pathway is refined while architectural parameters remain frozen and no MAC, parameter, or sensor penalties are applied. The effective objective is therefore

$$\mathcal{L}^{(1)} = \mathcal{L}_{task}, \tag{9}$$

which allows the backbone to learn a physiologically informative representation before compression pressures are introduced.

The second phase performs the joint search itself and uses a nominal budget of 70 epochs. In this stage, backbone and search parameters are jointly optimized with two optimizers, and the effective loss per batch becomes

$$\begin{aligned}\mathcal{L}^{(2)} = \mathcal{L}_{task} + \alpha_{eff}\mathcal{C}_{mac} + \beta_{eff}\mathcal{C}_{sen} \\ + \lambda_{par,eff}\mathcal{C}_{par} + \mathcal{P}_{min} + \beta_{eff}\mathcal{P}_{ent}\end{aligned} \tag{10}$$

The architectural optimizer uses a learning-rate multiplier of $10\times$ relative to the backbone updates, and compactness-oriented regularization may be delayed until a target fraction of the desired validation F_1 is reached, which helps preserve stability in severely data-constrained regimes.

The third phase is dedicated to discrete fine-tuning and uses a nominal budget of 20 epochs. After completion of the continuous search, the model activates a discrete forward path, freezes architectural parameters, and retrains only the backbone with the discretized channel mask and selected operations. The effective objective returns to

$$\mathcal{L}^{(3)} = \mathcal{L}_{task}. \tag{11}$$

Across all phases, progression is governed by validation Macro-F_1 with phase-wise early stopping (patience 15), checkpointing of the best phase state, and restoration of the corresponding weights before the next stage begins. This protocol limits overfitting, reduces unnecessary computation, and narrows the gap between the relaxed supernet and the final exportable model.

E. DISEASE-SPECIFIC MULTIOBJECTIVE MODEL SELECTION

The identification of the most appropriate architecture for each neurological disorder is formulated as a multiobjective selection procedure rather than a single-metric comparison. Candidate models are assessed jointly with respect to diagnostic discrimination, electrode economy, computational burden, temporal efficiency, and reproducibility across repeated evaluations. This formulation is motivated by the fact that clinically usable EEG systems must preserve predictive accuracy while remaining compatible with reduced montages and resource-constrained deployment.

Disease-specific model selection begins by acknowledging heterogeneity among cohorts, including differences in the number of available channels, class balance, and sample size. These factors influence the level of representational complexity that can be supported by the data and therefore require adaptive capacity assessment during model comparison. When capacity search is enabled, the framework

evaluates hidden dimensions in $\{32, 48, 64\}$ and numbers of search cells in $\{3, 4\}$, allowing the final recommendation to reflect both signal characteristics and data availability.

The decision process follows a constrained hierarchical strategy. First, candidate solutions with performance close to the best observed level are retained, i.e., those satisfying $F_1 \geq F_{1,\max} - \delta$. Within this admissible set, the recommended lexicographic ranking maximizes Macro-F_1, then minimizes sensor cost, MACs and parameter count, then maximizes stability across runs, and finally minimizes latency. When required for auditability, the same criteria may also be summarized through a normalized scalar score,

$$S = w_1 \hat{F}1 - w_2 \hat{C}_{sen} - w_3 \hat{C}_{mac} - w_4 \hat{C}_{lat} + w_5 \hat{E}_{stab}, \quad (12)$$

where hatted quantities are normalized to $[0, 1]$. The scalar score is auxiliary; the main decision remains grounded in hierarchical multi-criteria comparison.

To support the final recommendation for each disorder, candidate models are further subjected to complementary evidence procedures. These evidence blocks include component-removal analysis, robustness under corruption, reproducibility across repeated runs, and sensitivity to mask discretization. Together, these procedures are intended to support the preferred model not only by peak predictive performance but also by robustness, interpretability, and operational reliability.

A candidate is considered acceptable only when it exceeds the chosen reference baseline for the corresponding dataset, achieves at least 30% channel reduction without clinically meaningful loss of diagnostic performance, preserves a non-degenerate electrode configuration at the default threshold $\tau = 0.5$, remains more stable than a random-reference baseline, and satisfies the operational latency SLO defined for practical use. After these criteria are met, the selected continuous solution is converted into a compact discrete model and promoted as the preferred architecture for the target disorder. This designation should be interpreted as the best computational configuration within the experimental protocol, rather than as definitive clinical validation.

F. EXPERIMENTAL VALIDATION AND ROBUSTNESS ASSESSMENT

The methodological validation strategy is designed to demonstrate not only predictive effectiveness under ideal conditions, but also reliable behavior under realistic variations in EEG acquisition. In accordance with the project documentation, model assessment combines stratified partitioning of the available data for the main comparative tables, dedicated multi-seed runs for stability analysis, and a primary emphasis on the macro-averaged F1 score. This protocol reduces the risk of optimistic conclusions in small clinical cohorts and provides a more faithful estimate of generalization capacity.

Comparative evaluation is performed against complementary reference groups, including fixed architectures, models coupled with non-adaptive channel reduction, sequential optimization pipelines, and the proposed joint-search approach. This comparative structure makes it possible to distinguish gains attributable to joint structural adaptation from gains arising solely from feature reduction or model capacity. As a result, the experimental protocol supports a clearer interpretation of the contribution of each methodological component.

The broader validation framework is organized around four complementary evidence blocks: ablation, robustness under corruption, stability across repeated runs, and sensitivity to mask discretization. In the present manuscript, the reported evidence blocks cover ablation, stability, and fixed-threshold sensitivity ($\tau = 0.3, 0.5, 0.7$).

Statistical uncertainty is summarized through mean and standard deviation, complemented by 95% confidence intervals whenever the sampling regime allows reliable interval estimation. Matched comparisons by seed/fold are used whenever a direct contrast between candidate methods is required, thus preserving correspondence between evaluation conditions. This reporting strategy improves interpretability and prevents isolated high-performing runs from being mistaken as representative behavior.

Within the project validation framework, a configuration is considered methodologically robust when it surpasses the chosen reference baseline in macro-averaged F1 score, maintains stability above a random-reference baseline, preserves a non-degenerate channel subset after discretization, and, when corruption tests are run, exhibits predictable rather than erratic degradation. Taken together, these criteria ensure that the preferred model is supported by accuracy, reproducibility, and resilience, which are essential properties for translation from controlled experimentation to assistive EEG applications.

III. RESULTS AND DISCUSSION

A. BASELINE PERFORMANCE ACROSS DATASETS

Table 3 reports representative baseline results. Two consistent trends emerge. First, recurrent models are strongly competitive in depression and epilepsy, where GRU and BiGRU reach $F_1 = 0.880 \pm 0.037$ and 0.839 ± 0.022, respectively. Second, schizophrenia remains the hardest regime ($N = 84$), in which the best baseline is a statistical selector (variance-based, $F_1 = 0.621 \pm 0.102$), indicating that high-capacity temporal models are more sensitive to fold-level sampling variance under severe data scarcity.

TABLE 3. Cross-dataset baseline comparison (seed = 0, 5-fold CV).

Method	Depression	Schizophrenia	Epilepsy
All channels (fixed arch.)	0.544 ± 0.169	0.556 ± 0.218	0.713 ± 0.133
Variance selection	0.702 ± 0.283	0.621 ± 0.102	0.769 ± 0.040
NAS (fixed channels)	0.762 ± 0.078	0.607 ± 0.125	0.787 ± 0.084
Best EEG-DL per dataset	0.880 ± 0.037	0.464 ± 0.241	0.839 ± 0.022
Joint search (ours)	0.778 ± 0.075	$\mathbf{0.723 \pm 0.089}$	0.602 ± 0.074

Model-size analysis reinforces a complexity–generalization trade-off. In depression, a compact GRU (20,738 parameters)

outperformed larger BiGRU and attention-enhanced variants (41,282 and 45,442 parameters), indicating that increased capacity can amplify variance when the per-channel feature axis is compact ($T = 20$) and N is limited. In epilepsy, where $N = 318$, larger recurrent capacity became beneficial, consistent with the lower fold-to-fold variability observed in Table 3.

The differentiable-mask-only baseline exhibited a recurrent collapse to 0–2 active channels ($F_1 \approx 0.33$–0.40 across datasets), confirming that standalone ℓ_1-dominated mask optimization is unstable in low-data settings and motivating the regularized joint formulation.

B. JOINT CHANNEL-AND-ARCHITECTURE OPTIMIZATION

The core joint-search outcomes are summarized in Table 4. Across all disorders, the model selected only 4–6 channels, corresponding to a 68–71% reduction in electrodes. This reduction is clinically relevant because electrode count directly affects setup time, gel usage, and patient comfort during ambulatory acquisition.

TABLE 4. Joint-search performance and channel reduction.

Dataset	Joint F_1	Best-fold F_1	Channels	Reduction
Depression	0.778 ± 0.075	0.873	6/19	68%
Schizophrenia	0.723 ± 0.089	0.821	5/16	69%
Epilepsy	0.602 ± 0.074	0.752	4/14	71%

To contextualize these results, Table 5 compares the joint model with a sequential channel-then-architecture baseline. Although the sequential approach is numerically higher in all datasets, the depression score (1.000 ± 0.000) under 5-fold validation with approximately 24 validation samples per fold should be interpreted cautiously and warrants a dedicated leakage audit plus repeated-run verification. The joint objective, while less optimistic numerically, behaves as an implicit regularizer and avoids similarly degenerate operating points in the reference run.

TABLE 5. Joint versus sequential optimization.

Dataset	Joint F_1	Sequential F_1	Δ (Joint–Seq.)
Depression	0.778 ± 0.075	1.000 ± 0.000	-0.222
Schizophrenia	0.723 ± 0.089	0.818 ± 0.113	-0.095
Epilepsy	0.602 ± 0.074	0.737 ± 0.055	-0.135

The epilepsy gap ($\Delta = -0.135$) is more informative than the depression gap and suggests a genuine optimization burden: jointly optimizing architecture and channel masks creates a harder loss landscape than two-stage pipelines when model capacity is high.

C. STABILITY AND THRESHOLD SENSITIVITY

Table 6 reports the stability block of the validation framework, summarized here by mean Jaccard overlap between channel masks and operation consistency per search cell across repeated runs. Mean Jaccard values are above the reported random-reference values, suggesting that channel importance is not random across seeds. In contrast, operation consistency is dataset-dependent and is highest in epilepsy.

TABLE 6. Stability analysis across independent runs.

Dataset	Mean Jaccard	Random Jaccard	Cell-op consistency
Depression	0.320	$\sim$0.16	0.000
Schizophrenia	0.289	$\sim$0.16	0.000
Epilepsy	0.294	$\sim$0.16	0.250

This dissociation between stable channel masks and unstable architecture choices indicates that spatially informative electrodes can be identified under low N, whereas architecture ranking is less consistent in the smaller cohorts studied here. In the present study, the most consistent architecture-level signal was observed in the largest cohort ($N = 318$), but broader validation across additional datasets is still needed.

Table 7 reports the fixed-threshold portion of the discretization-sensitivity analysis. The default deployment threshold is $\tau = 0.50$, which is also the best setting for depression and epilepsy in these runs. For schizophrenia, $\tau = 0.30$ (retaining all channels) yields the highest unconstrained F_1; however, that operating point removes the sensor-efficiency benefit, so the reported 5/16-channel result in Table 4 reflects the default multiobjective deployment rule at $\tau = 0.50$ rather than the purely F_1-maximizing threshold.

TABLE 7. Threshold sensitivity (F_1 with number of selected channels in parentheses).

Dataset	$\tau = 0.30$	$\tau = 0.50$	$\tau = 0.70$	Best τ
Depression	0.333 (19)	0.873 (6)	0.333 (0)	0.50
Schizophrenia	0.764 (16)	0.702 (5)	0.346 (0)	0.30
Epilepsy	0.425 (14)	0.538 (4)	0.360 (0)	0.50

The systematic failure at $\tau = 0.70$ (all channels removed) indicates that learned mask probabilities remain mostly in the mid-sigmoid range under regularization.

D. ABLATION STUDY AND TRADE-OFF INTERPRETATION

Table 8 isolates each component contribution. The active-channel counts reported in this table correspond to the mean discretized counts within the ablation runs and therefore should not be interpreted as the final deployment montage summarized in Table 4. The most consistent observation is that removing channel selection yields the highest F_1 in both datasets reported, but only by restoring all electrodes.

These ablations support two complementary interpretations. First, mask and NAS gradients can interfere during joint training, reducing pure predictive accuracy. Second, the methodological target is not unconstrained F_1 maximization; instead, it is Pareto-efficient operation under acquisition constraints. From this perspective, sacrificing part of the

TABLE 8. Ablation study under the same 5-fold protocol.

Dataset	Configuration	Mean F_1	Active channels
Depression	full_model	0.621±0.263	3
Depression	no_sensor_cost	0.582±0.231	12
Depression	no_hw_cost	0.685±0.207	4
Depression	no_channel_selection	**0.958±0.029**	19
Depression	no_nas	0.687±0.219	4
Depression	reduced_search	0.612±0.240	4
Epilepsy	full_model	0.586±0.075	2
Epilepsy	no_sensor_cost	0.685±0.053	7
Epilepsy	no_hw_cost	0.533±0.118	3
Epilepsy	no_channel_selection	**0.898±0.043**	14
Epilepsy	no_nas	0.633±0.046	2
Epilepsy	reduced_search	0.539±0.057	3

maximum achievable F_1 to obtain a substantially smaller electrode set is an intended design outcome rather than a failure mode.

E. EMBEDDED DEPLOYMENT PROFILE

Table 9 reports deployment-oriented characteristics of the resulting models. Under the project deployment framework, latency is treated as an operational SLO and part of the promotion criteria. All variants remain near 260k parameters and satisfy the low-memory deployment target emphasized by the framework ($\leq$2 MB flash). The reported FP32 inference latency ranges from 4.46 to 7.02 ms.

TABLE 9. Embedded deployment characteristics of joint-search models.

Dataset	Params	FP32 (KB)	INT8 (KB)	FP32 latency (ms)	Channels
Depression	∼260k	1073	∼266	7.02	6/19
Schizophrenia	∼260k	1073	∼266	4.48	5/16
Epilepsy	∼260k	1072	∼266	4.46	4/14

The current dynamic-quantization pipeline did not fully exploit theoretical compression limits. Post-training static quantization with calibration data is therefore a practical next step to improve memory reduction without retraining.

F. SYNTHESIS AND LIMITATIONS

Overall, the proposed framework delivers a robust sensor-efficiency gain with clinically meaningful accuracy, particularly in schizophrenia where joint optimization establishes the best cross-method result under low N. The central contribution is therefore a controllable performance–efficiency frontier, not absolute F_1 maximization.

The primary limitation is that the main cross-dataset performance comparison is still anchored to a reference run, while the stability analysis is reported separately across repeated random initializations. Ongoing broader multi-seed experiments are required to replace fold-based uncertainty with full run-to-run confidence intervals for every comparison block. In addition, operation consistency remains limited in the two smallest datasets, indicating that future work should combine reduced search spaces with bilevel optimization and stronger priors for low-sample regimes.

G. CRITICAL CONSISTENCY ASSESSMENT

The new results both reinforce and challenge the initial hypotheses, and this dual outcome is scientifically informative rather than contradictory. First, the findings reinforce the main translational claim: as shown in Table 4, the method consistently achieves large channel reduction (68–71%) while maintaining usable diagnostic performance, and Table 9 reports memory and latency values aligned with the low-burden deployment objective. These results support the practical objective of low-burden EEG acquisition.

Second, the results challenge any strong claim that joint optimization universally maximizes predictive accuracy. Table 8 shows that, in the reported ablations, removing channel selection (i.e., using all electrodes) yields the highest F_1 in depression and epilepsy, and Table 5 shows higher sequential scores across datasets. However, this apparent contradiction is resolved by the optimization target itself: the proposed formulation is multiobjective and intentionally penalizes sensor and hardware cost. Therefore, lower unconstrained F_1 relative to all-channel configurations is expected at the Pareto-efficient operating point.

Third, operation-level stability partially challenges the notion of uniformly reliable architecture discovery in small cohorts. Table 6 indicates that channel-selection stability is consistently above random, whereas operation consistency remains low in depression and schizophrenia. This suggests that, under small N, the framework is more reliable for identifying informative spatial subsets than for achieving consistent operation selection.

Taken together, these results refine the paper's interpretation: the framework should be viewed as a principled trade-off mechanism for sensor-efficient deployment, with strongest evidence for channel prioritization and more limited evidence for architecture convergence in the reported datasets.

IV. CONCLUSION

This work presented a joint channel-and-architecture optimization framework for EEG classification in depression, schizophrenia, and epilepsy under realistic clinical data constraints. Across all datasets, the method consistently reduced electrode usage by 68–71%, reaching compact configurations of only 4–6 channels while preserving competitive macro-F_1.

The empirical analysis demonstrates that the approach is particularly effective in the schizophrenia setting, where it outperformed reference deep baselines despite severe sample scarcity. For depression and epilepsy, the results explicitly reveal a controlled contradiction: maximum predictive accuracy is obtained when all channels are retained, whereas joint optimization improves acquisition efficiency at the cost of part of the attainable F_1. This trade-off is consistent with the multi-objective formulation rather than a methodological failure.

Stability and ablation analyses indicate that channel selection is more consistent than architecture selection in the reported runs. These findings suggest a practical guideline

for translational use: in small cohorts, joint search is most reliable as a channel-prioritization mechanism, while larger cohorts better support full architecture adaptation. Overall, the proposed framework advances EEG-NAS toward more compact, interpretable, and hardware-efficient diagnostic systems.

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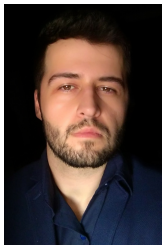
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